A Deep Learning Framework on Macular OCT for Automated Multi-Attribute Metadata Recovery for Large-Scale Ophthalmic Biobanks

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Abstract. Accurate metadata annotation is critical for large-scale Optical Coherence Tomography (OCT) datasets, yet patient attributes such as age, sex, and eye laterality are frequently missing or inconsistent in clinical databases. We propose a deep learning framework for automated metadata recovery from macular OCT B-scans, evaluating two distinct strategies: (i) a Task-Specific Framework (TSF) utilizing specialized models, including Swin Transformer-Base and hybrid DenseNet-Transformer architectures, combined via late fusion, and (ii) a Unified Multi-Head Framework (MHF) that jointly predicts all attributes from a shared ResNet-50 backbone. Our experiments on a mixed clinical dataset demonstrate that the optimized single-task Swin-B model achieves a state-of-the-art age estimation MAE of 4 years. Furthermore, the framework reached 90% accuracy for sex prediction and over 93% for eye laterality, maintaining robust performance across various retinal pathologies including AMD and DME. While the TSF achieves the highest predictive accuracy, the unified MHF offers a significantly more compact alternative despite slight performance trade-offs. These findings confirm that macular OCT images encode rich latent demographic and anatomical information, enabling effective automated metadata recovery for large-scale biobank curation.

Keywords: Optical Coherence Tomography (OCT) · Metadata Recovery · Deep Learning · Single-Task vs Multi-Task Learning · Vision Transformers

1 Introduction

Optical Coherence Tomography (OCT) has revolutionized ophthalmic care by providing non-invasive, high-resolution cross-sectional visualizations of retinal morphology. While its primary clinical utility lies in the diagnosis and longitudinal monitoring of pathologies such as age-related macular degeneration and glaucoma, recent advancements in deep learning (DL) have revealed that OCT B-scans contain a wealth of "hidden" biological signals. Beyond overt structural lesions, these images encode fundamental demographic and physiological metadata, including age, biological sex, and eye laterality, that are often imperceptible to human graders. The automated recovery of this metadata, a process termed "metadata recovery," is increasingly critical for large-scale retrospective research. High-volume clinical registries often suffer from "metadata sparsity," where missing labels or inconsistent Electronic Health Records (EHR) limit the utility of the data for population-level analysis. Despite the potential, current approaches to metadata extraction remain fragmented. Most existing models are designed as single-task architectures, focusing on a single attribute within curated, healthy cohorts. Moreover, the robustness of these "latent signal" extractors in the presence of ocular disease remains under-explored.

In this work, we propose a deep learning approach for automated metadata recovery for the recovery of age, sex, and eye laterality from raw OCT B-scans. We systematically compare a Task-Specific Framework (TSF) utilizing specialized models against a unified Multi-Head Framework (MHF) to evaluate the trade-offs between feature specialization and shared representation learning. By validating our framework on a diverse dataset comprising both healthy and diseased eyes, we demonstrate a robust solution for automating OCT metadata extraction. The remainder of this paper is structured as follows. Section 2 synthesizes existing literature on OCT-based metadata prediction. Section 3 presents our proposed methodology, detailing the architectural design of both the standalone single-task baselines and the unified multi-head framework. Section 4 provides a comprehensive empirical evaluation, including comparative analyses against state-of-the-art. Finally, Section 5 summarizes our key findings and discusses prospective directions for deep learning-based metadata recovery.

2 Related Work

Deep learning has become increasingly prominent in ophthalmic imaging, with OCT providing high-resolution structural information not only for disease diagnosis but also for demographic and anatomical prediction tasks. Age prediction from OCT has been widely studied as a proxy for biological aging and retinal microstructural integrity. The study [12] first demonstrated that chronological age can be estimated from OCT scans using CNNs, reporting MAE values of approximately 6–7 years on largely healthy cohorts. Building on this, work in [3] leveraged the large-scale UK Biobank OCT dataset to estimate an "OCT age," revealing strong correlations with systemic health indicators and highlighting the retina as a marker of whole-body aging. More recently, the study [11] showed that combining structural OCT with OCT angiography features improves age prediction accuracy, underscoring the complementary role of vascular information. Beyond age estimation, multiple studies have demonstrated that OCT encodes sex-specific retinal patterns. In [4] the authors reported sex classification accuracies of approximately 85% from macular OCT volumes while jointly predicting age, while in [9]showed that optic nerve head scans can be used to infer sex and age across different retinal regions. Additional work using anterior segment OCT further confirms the feasibility of demographic prediction from ocular structures [2,7]. Eye laterality prediction has also been explored as a metadata integrity task. The study in [10] demonstrated that deep learning models can accurately distinguish left from right macular OCT scans, achieving accuracies exceeding 95% despite subtle anatomical asymmetries. More recently, metadata has been incorporated into self-supervised and contrastive learning frameworks. Authors in [5] introduced metadata-enhanced contrastive learning that improves downstream OCT performance, while in [13,6] was showed that multi-stage self-supervised pretraining on mixed OCT datasets enhances diagnostic accuracy. Despite these advances, most prior studies focus on individual metadata attributes or specific OCT regions and do not address joint metadata prediction within a unified framework. Moreover, practical solutions for metadata reconstruction remain limited, despite the prevalence of incomplete labeling in large OCT repositories. To address these gaps, this work proposes a A Multi-Task Deep Learning framework for automated prediction of age, sex, and eye laterality from macular OCT scans, evaluated on mixed clinical datasets containing both healthy and diseased cases.

3 Methodology

This section presents a robust deep learning pipeline for metadata recovery from macular OCT imaging. To evaluate the efficacy of shared representation learning, our methodology follows two-phase design: (1) task-specific architectures, and (2) a unified multi-head framework designed for multi-attribute prediction. We further describe the data curation process, the backbone selection, and the training strategies employed to balance the competing objectives of age regression and categorical classification of sex and laterality.

3.1 Dataset Overview

General dataset description Experiments used the publicly available **OCTDL v4** dataset, which contains 2,064 macular OCT B-scans annotated for retinal pathologies, including age-related macular degeneration (AMD), diabetic macular edema (DME), epiretinal membrane (ERM), retinal artery occlusion (RAO), retinal vein occlusion (RVO), vitreomacular interface disease (VID), and normal eyes (NO) [1]. All images were fovea-centered, acquired under standardized protocols, and reviewed by retinal specialists. Patient metadata (age, sex, and eye laterality) served as prediction targets and were linked via structured filenames. Only samples with available ground-truth labels were used for task-specific training. The dataset is publicly accessible in [1]. The dataset exhibits substantial missing metadata, with age, sex, and eye laterality unspecified for 78% of scans. In terms of disease distribution, the majority of scans correspond to AMD (1,231 images), followed by normal eyes (NO, 332 images). Other pathologies include ERM (155 images), DME (147 images), RVO (101 images), VID (76 images), and RAO (22 images), highlighting the predominance of AMD in the dataset, while rarer conditions such as RAO are underrepresented.

Metadata description The metedata exhibits imbalance across predicted attributes. Among labeled samples (n = 448):

- **Age:** ranges from 19 to 92 years (mean: 66, std: 14), with the majority of samples between 60–80 years.

- **Sex:** includes 125 male and 323 female samples (27.9% vs. 72.1%), reflecting moderate imbalance.
- **Eye laterality:** 203 right-eye (Oculus Dexter, OD) and 245 left-eye (Oculus Sinister, OS) scans (45.3% vs. 54.7%), showing mild imbalance.

These reduced effective sample sizes may introduce selection bias, as metadata availability could correlate with disease type or acquisition conditions, potentially affecting generalization. Task-specific balancing was applied during training, and minority classes were addressed using random oversampling to mitigate the effects of this imbalance. We note this limitation and interpret prediction performance with caution.

Selection Bias and Representativeness To assess potential bias, we examined the relationship between metadata availability (age, sex, eye) and disease classification. A Chi-square test revealed a highly significant association between disease type and metadata availability ($p \approx 2.32 \times 10^{-18}$), indicating that missingness is not random.

Metadata completion rates vary markedly across disease categories. For instance, AMD exhibits the highest availability 28%, whereas RAO shows 0% availability and DME only 2.7% availability. This systematic discrepancy confirms a disease-dependent pattern of missingness.

Within the subset records containing demographic information, age differs substantially across diseases (AMD: 71.2 $\pm$ 10.4 years; NO: 50.5 $\pm$ 17.4 years). No statistically significant association was found between sex and disease ($p = 0.156$), and only a marginal trend was observed for eye laterality ($p = 0.060$).

Collectively, these findings indicate that metadata is Missing Not At Random (MNAR), with availability strongly conditioned on disease category. As a result, demographic analyses are inherently restricted to a group with certain diseases. Nevertheless, by explicitly quantifying these dependencies, the study ensures transparent interpretation of model performance within the analyzed population.

3.2 Architectural Design 1 : Task-Specific Framework (TSF)

The TSF employs three independent deep learning models, each optimized specifically for age, sex, or eye laterality. These standalone predictions are integrated via decision-level fusion. Each task is described as follows:

A. Task 1 - Age Regression: A continuous prediction aimed at estimating chronological age from retinal patterns. Aging manifests in OCT scans through subtle, non-local changes, including retinal layer thinning, foveal contour shifts, and the accumulation of drusen or deposits. To capture these spatially distributed features, we utilize the *Swin Transformer-Base (Swin-B)* backbone [8]. The hierarchical shifted-window self-attention of *Swin-B* is uniquely suited for this task: early layers capture fine-grained textures, while deeper layers encode the global anatomical context required for robust age estimation (Fig. 1A). The model is initialized with *ImageNet-pretrained* weights, with the original classification head replaced by a regression block, as shown in Fig. 1B. *Swin-B*'s multi-scale feature extraction and efficient handling of high-resolution images

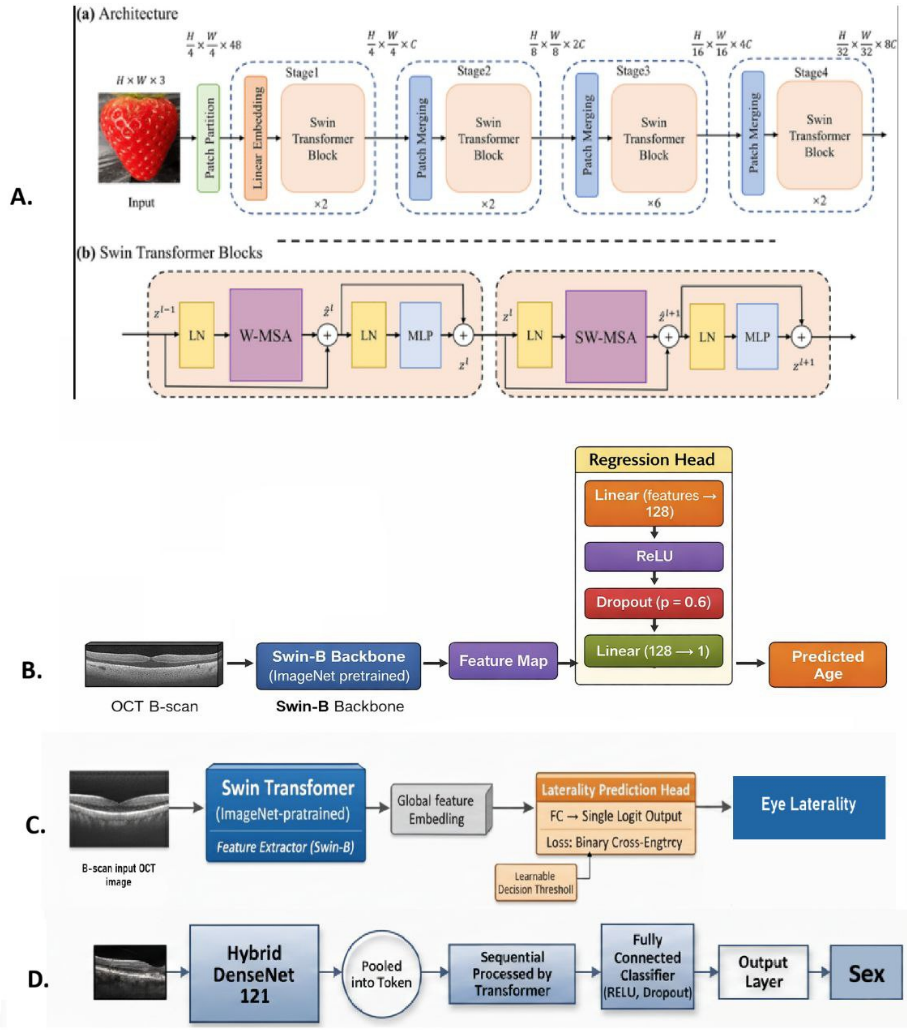


Fig. 1: Architectural Overview of Task-Specific Frameworks (TSF). **(A)** The Swin Transformer-Base (Swin-B) backbone [8]. **(B)** Task 1 : Age Regression. **(C)** Task 2: Eye Laterality Identification. **(D)** Task 3: Sex Classification.

enable the detection of distributed retinal biomarkers that are often imperceptible to human graders.

B. Task 2 - Eye Laterality Identification: A binary classification to distinguish left (OS) from right (OD) eyes. We employ the *Swin Transformer–Base (Swin-B)* as the feature extraction backbone (Fig. 1C), utilizing a single-logit output with a learnable threshold to mitigate class imbalance. To preserve essential laterality cues, we apply orientation-preserving augmentations while strictly avoiding horizontal flips. *Swin-B* is particularly effective for this task as laterality depends on orientation-specific retinal features, including asymmetries in foveal positioning and vascular patterns. Its hierarchical shifted-window self-attention integrates local textures with global structures, allowing the model to detect spatially distributed cues in high-resolution scans.

C. Task 3 - Sex Classification: A binary classification to determine biological sex from subtle retinal morphological differences. We employ a *hybrid DenseNet-121 + Transformer* architecture (Fig. 1 D). This design leverages *DenseNet-121*'s dense blocks to facilitate feature reuse and preserve detailed retinal textures. These extracted features are then pooled, tokenized, and processed by a lightweight *Transformer encoder* with multi-head self-attention to capture long-range dependencies and global structural context. Class imbalance is addressed via oversampling. This hybrid approach is particularly effective as it integrates fine-grained local textures with global dependencies, enabling the robust extraction of the subtle, distributed retinal features associated with biological sex.

D. Wrapper-Based Late Fusion For multi-task predictions, independently trained models for age, sex, and eye laterality are combined via decision-level, wrapper-based fusion. Each task-specific model M_{task} processes the input scan x to generate an individual prediction $\hat{y}_{task} = M_{task}(x)$. These are then aggregated into a unified inference vector: $\hat{y} = \left[\hat{y}_{\text{age}}, \hat{y}_{\text{sex}}, \hat{y}_{\text{eye}}\right]$. This strategy enables a unified multi-task inference framework while preserving task-specific feature specialization. By decoupling the training processes, we ensure that the optimization of one attribute (e.g., age regression) does not interfere with the feature extraction of another (e.g., laterality classification), providing a robust baseline for multi-label metadata recovery.

3.3 Architectural Design 2: Multi-Head Framework (MHF)

We propose a unified deep learning model for simultaneous prediction of age, sex, and eye laterality from macular OCT scans. A shared backbone learns common retinal features, followed by three task-specific heads: age regression, sex classification, and eye laterality classification (Fig. 2). Age uses a linear output, while sex and eye laterality use softmax. Several backbone architectures were explored within the unified multi-head framework, including VGG16, DenseNet-121, Vision Transformer (ViT-B/16), Swin Transformer, and ResNet-50, all initialized with ImageNet pretrained weights and fine-tuned on OCT images. For ResNet-50, the first convolutional layer was adapted to accept single-channel OCT inputs, and the final fully connected layer was replaced with an identity mapping to extract feature embeddings. Transformer based backbones (ViT and Swin) were used with their standard configurations. In all cases, the extracted feature representations were forwarded to task-specific prediction heads. Based on comparative validation performance, ResNet-50 was selected as the final backbone (Fig. 2) due to its superior and more balanced accuracy across age, sex, and eye laterality prediction tasks, as well as its stable training behavior on the mixed clinical dataset.

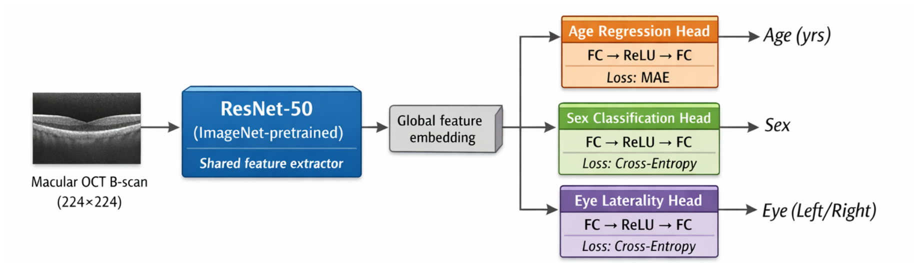


Fig. 2: Architectural Overview of Multi-Head Framework (MHF) showing a Unified ResNet-50 Backbone for Simultaneous Metadata Predictions.

4 Results and Discussion

This section presents a comprehensive evaluation of the proposed deep learning frameworks for metadata recovery from macular OCT B-scans. We begin by analyzing the performance of the *Task-Specific Framework (TSF)*. Subsequently, we evaluate the *Multi-Head Framework (MHF)*, comparing their efficiency and predictive accuracy against the standalone models and current state-of-the-art methods. The data was split into training (60%), validation (20%), and test (20%) for all the models.

4.1 Task-Specific Framework (TSF)

A. Age Regression: Age prediction was performed on 448 OCTDL B-scan images, z-score normalized using training-set statistics, with performance measured via Mean Absolute Error (MAE). Multiple CNN and transformer backbones were compared in Table 1. Transformer-based models outperformed CNNs, with Swin Transformer Base (Swin-B) achieving the lowest MAE due to hierarchical feature learning and windowed self-attention. DenseNet-121 provided a strong CNN baseline (6.65), while EfficientNet (7.50), ResNet (7.72), and ViT-B/16 (7.53) were less effective. As detailed in Table 2, three Swin-B variants were evaluated to determine the impact of augmentation strength and optimization configurations on age estimation. Runs 1 and 3 utilized moderate augmentations (horizontal flips, 10° rotations, and ColorJitter), while Run 2 implemented stronger augmentations, including vertical flips, 20° rotations, and Gaussian blur. Results indicate that excessive data transformation in Run 2 led to a higher MAE (6.29), suggesting that aggressive distortions may obscure subtle retinal features critical for age regression. Conversely, Run 3 achieved the best performance (MAE = 4.27) by pairing moderate augmentations, Automatic Mixed Precision (AMP) with robust regularization, including 0.6 dropout and AdamW optimization (0.1 weight decay). This highlights that a balanced approach to geometric transformations is essential for accurate OCT-based age estimation.

B. Eye Laterality Identification: Eye laterality was classified on 448 OCT images using DenseNet-121 + Transformer, ResNet18/50, and Swin-B Transformer as shown in Table 3. Transformer based models outperformed CNNs, with Swin-B achieving the highest metrics of Accuracy, ROC-AUC and F1-Score over 93% by leveraging hierarchical attention to capture both local and global retinal features. CNNs captured local textures effectively, but Swin-B's hierarchical self

Table 1: Performance comparison of different Backbones for age regression

Model Architecture	Test MAE (Years)
DenseNet-121	6.65
EfficientNet	7.50
ResNet	7.72
ViT-B/16	7.53
Swin-B (Optimized)	**4.81**

Table 2: Training Strategies for the *Swin-B* Architecture.

Variant	AMP	Dropout	Scheduler	Augmentation	Test MAE
Swin-B: Run1	No	0.5	Cosine	Moderate	5.50
Swin-B: Run2	Yes	0.6	ReduceLROnPlateau	Strong	6.29
Swin-B: Run3	**Yes**	**0.6**	**ReduceLROnPlateau**	**Moderate**	**4.27**

attention and windowed global context modeling provided superior performance. Training used rotations ($\pm 10°$), color jitter, resized cropping, Adam optimizer (LR$=10^{-4}$, weight decay$=10^{-4}$), binary cross-entropy, 30 epochs, batch size 8, mixed-precision training, and a learnable threshold.

Table 3: Performance comparison of different Backbones for Eye laterality.

Model	Accuracy	ROC-AUC	F1-Macro
DenseNet-121 + Transformer	0.6701	0.7568	0.66
ResNet18	0.7667	0.9097	0.76
Swin-B Transformer	**0.9333**	**0.9351**	**0.93**

C. Sex Classification: We predict biological sex (Male/Female) from 448 OCT images using a Hybrid DenseNet-121 + Transformer. The dataset was balanced via oversampling and included flips, rotations, and color jittering for augmentation. The model uses a frozen ImageNet pretrained DenseNet-121 backbone for local feature extraction, followed by adaptive pooling (14×14), a 2-layer Transformer encoder with 8 attention heads, and a two layer MLP with dropout (0.4) and ReLU producing the final prediction. This architecture captures both fine grained retinal textures and long range correlations. The hybrid model outperforms alternatives by combining DenseNet's dense local feature extraction with the Transformer's global context modeling, which is particularly effective for small datasets with subtle retinal patterns.

Table 4: Performance comparison of different Backbones for Sex classification.

Model	Accuracy	AUC	F1
DenseNet-121 + Transformer	**0.8462**	**0.9128**	**0.852**
DenseNet-121	0.7923	0.8715	0.803
ResNet-50	0.7692	0.8452	0.781
VGG-16	0.7231	0.7983	0.728
Simple CNN	0.6538	0.7126	0.667
ViT-Base	0.6846	0.7451	0.692

D. Late Fusion Wrapper: The late fusion stage integrates independently pretrained age, sex, and eye laterality models by forwarding each OCT B-scan through frozen networks. As shown in Table 5, on a shared test set, the framework demonstrates high predictive stability across various retinal diseases. Age prediction achieved low error rates across the board, however, interpretation depends on sample size. Very small classes, such as DME (n = 4, MAE = 0.140), may exhibit low errors that are not necessarily indicative of meaningful disease-related patterns. In contrast, larger classes like AMD (n = 348, MAE = 0.291) show more substantial variation, suggesting that factors beyond sample size including disease heterogeneity influence regression performance. More challenging conditions, such as Normal (n = 48, MAE = 0.483) and ERM (n = 22, MAE = and 0.509), yielded higher error margins, further supporting this observation. Sex classification remained robust, with the highest accuracy observed in DME and Vitreous Impairment/Detachment (VID) (100%); performance was slightly lower for Age-Related Macular Degeneration (AMD) (91.7%) and Retinal Vein Occlusion (RVO) (87.5%). Eye laterality identification reached near-perfect accuracy across most categories, with RVO and VID achieving 1.000. The lower laterality accuracy for DME (0.750) suggests that severe fluid accumulation may occasionally obscure the anatomical markers used for OS/OD differentiation.

Table 5: Performance of the Late Fusion Wrapper across different retinal diseases.

Disease	**Samples**	**Age MAE**	**Age STD**	**Sex Acc**	**Eye Acc**
DME	4	0.140	0.069	1	0.750
AMD	348	0.291	0.297	0.917	0.945
RVO	8	0.295	0.189	0.875	1.000
VID	18	0.412	0.252	1	1.000
NO	48	0.483	0.461	0.979	0.896
ERM	22	0.509	0.384	0.955	0.955

4.2 Multi-Head Framework (MHF)

Unified multi-head models were evaluated for simultaneous prediction of age, sex, and eye laterality using ResNet-50, Swin-Tiny Transformer, VGG16, DenseNet-121, and ViT-B/16. Age performance was measured by MAE, while sex and eye laterality were assessed using accuracy (Table 6). ResNet-50 offered the most balanced performance across tasks, achieving moderate age MAE with strong sex and eye laterality accuracies. Swin-Tiny excelled in age prediction due to its hierarchical attention but performed worse on categorical tasks. DenseNet-121 benefited from efficient feature reuse but underperformed in eye laterality, while VGG16 and ViT-B/16 showed weaker overall results. Classification metrics for ResNet-50, showed in Table 7, indicate consistent precision and recall for sex (78.5% and 77.2%) and eye laterality (76.8% and 75.4%). The ResNet-50 unified model was trained end-to-end using the Adam optimizer with a learning rate of 1×10^{-4}, a batch size of 16, and for 25 epochs.

Table 6: Performance comparison of different backbones within the MHF.

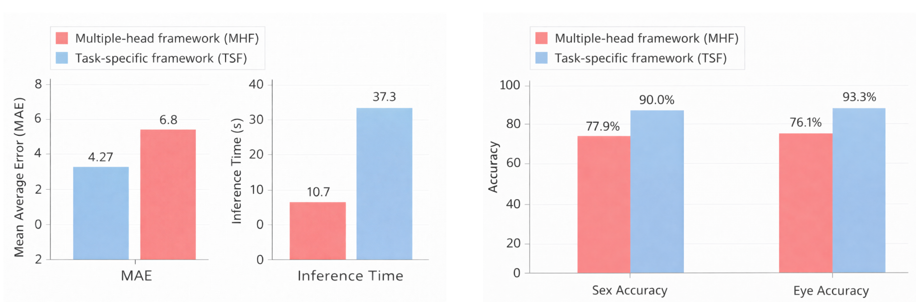
Backbone Model	Age MAE (years)	Sex Accuracy	Eye Accuracy
ResNet-50	6.8	**0.779**	**0.761**
Swin-Tiny Transformer	**6.45**	0.671	0.72
VGG16	7.5	0.657	0.69
DenseNet-121	7.2	0.717	0.67
ViT-B/16	6.65	0.650	0.680

Table 7: Performance metrics for metadata prediction. Sex and eye prediction results are obtained from the ResNet-50 multi-head model, while age prediction (MAE) is obtained from the Swin-Tiny model.

Metric	Eye Prediction	Sex Prediction	Age Prediction
Accuracy	0.761	0.779	-
Precision	0.768	0.785	-
Recall	0.754	0.772	-
F1-score	0.761	0.778	-
Mean Absolute Error	-	-	6.45

4.3 Comparative Study

1. Task-Specific vs Multi-Head Frameworks: Our evaluation shows that single-task models generally outperform the unified multi-task framework across all attributes . This gap is largely due to gradient conflicts arising from the joint optimization of heterogeneous tasks, where age regression and classification of sex and eye laterality produce competing gradients that hinder convergence and task-specific feature learning. In terms of model complexity, the single-task (Late Fusion) framework has a substantially larger parameter footprint (Table 8), allowing each model to learn highly specialized representations at the cost of higher memory usage and longer inference time. We note that this comparison reflects not only differences in learning paradigm but also model capacity, as TSF employs larger transformer-based backbones (e.g., Swin-B) while the MHF uses ResNet-50 to prioritize efficiency. The unified framework therefore emphasizes compactness and deployability rather than maximizing raw predictive accuracy.



(a) Sex and Eye Accuracy Comparison (b) MAE and Inference Time Comparison

Fig. 3: Comparison between Task-Specific Framework (TSF) and Multiple-Head Framework (MHF) in terms of predictive performance and computational efficiency.

Table 8: Comparison of parameter counts and per-image inference time between Task-Specific and Multi-Head Frameworks

Model Framework	**Total Params**	**Trainable Params**	**Inference Time (ms)**
Multi-Head	25,557,032	25,557,032	10.7
Task-Specific	182,947,421	175,993,565	37.3

2. Performance Against State-of-the-Art: Our optimized single-task Swin-B model achieves an age prediction MAE of 4.27 years, surpassing previous benchmarks set in [12] (approx. 5–7 years) and in [3] (approx. 5–6 years). For sex prediction, our framework reaches an accuracy of 90.0%, outperforming the 85% reported in [4]. Furthermore, eye laterality classification achieves 93.33%, which is highly competitive with the state-of-the-art performance reported in [10]. These results collectively demonstrate that our proposed specialized frameworks set a new benchmark for automated metadata recovery from macular OCT scans.

5 Conclusion

This study demonstrates that macular OCT B-scans contain rich demographic and anatomical signals, effectively captured by deep learning for the automated prediction of age, sex, and eye laterality. While the unified multi-head framework offers significant parameter efficiency and lower inference costs, specialized single-task models, particularly Swin Transformer for age and laterality and convolutional/hybrid models for sex, yield superior accuracy by avoiding the gradient conflicts inherent in joint optimization. Despite limitations in dataset size and task-balancing strategies, training on mixed clinical data improved the model's robustness and generalization across various retinal pathologies. These findings suggest that automated metadata recovery can significantly enhance large-scale clinical database curation by recovering missing labels, with future work focused on utilizing adaptive loss weighting and self-supervised pre-training to better balance efficiency with predictive performance. **Furthermore, the evaluation was conducted on a single dataset, which may limit generalizability to other imaging devices, populations, or acquisition settings. Performance on disease categories with very small sample sizes should be interpreted cautiously, and future work will include cross-dataset validation to better assess robustness and external validity.**

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